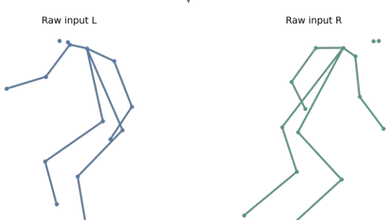
System overview of CanonFuse3D

A Two monocular 3D pose streams

Upstream RGB: view L / view R

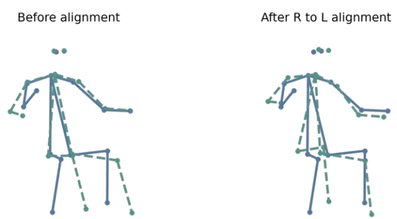


Monocular 3D estimation



P^L P^R

B Human-centric coordinates + pose alignment



Pose-space alignment
R → L and L → R

C Cross-view + temporal features



Five cues · 15 channels
per joint and frame

D Joint-frame gate + residual correction

$$Z = [H^L, H^R]$$

$$\alpha = \sigma(g_\theta(Z))$$

$$\Delta P = h_\theta(Z)$$

$$\alpha B^L + (1 - \alpha)B^R + \Delta P$$

E Refined canonical 3D pose sequence

Refined canonical output



P^F

Approximately synchronized 3D pose inputs; no camera parameters at the fusion stage.

Self-collected outdoor example · A01 / pro_1 / frame 746 · Qualitative illustration; no ground truth